

MATRIXPOLICY FOR ROLE-AWARE OPTIMIZATION OF RATIONAL TRANSFORMER NONLINEARITIES

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ABSTRACT

A useful LLM pretraining recipe moves the loss-time frontier by reaching the same validation loss with fewer tokens or less wall-clock time. We target this objective through the Transformer nonlinear sublayer, treating it as an optimizer-visible interface. MatrixPolicy with Rational Latent Basis (RLB) replaces a fixed scalar response with grouped rational responses that expose local gain, coefficient-activity, and matrix-balance statistics to a role-aware update. MatrixPolicy restricts the signal-conditioned update to the sublayer input and output matrices; rational coefficients and the rest of the Transformer remain on AdamW. Across DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4 with a fixed 12-layer, width-768 decoder, MatrixPolicy reaches matched validation-loss targets with fewer tokens and lower measured arrival time than SiLU controls under the same protocol. It saves 19.5–22.8% target-arrival tokens versus SiLU with AdamW and 29.2–34.0% versus SiLU with Muon at 100M tokens, and 22.3–26.4% and 6.7–9.8%, respectively, at 300M tokens. [todo: Insert the larger-scale training result here.]

1 INTRODUCTION

Recent LLM optimizer papers frame practical efficiency through loss-time and compute-time frontiers, asking which training recipe reaches a target loss with fewer tokens, FLOPs, or wall-clock time (Shah et al., 2025; Liu et al., 2025; Semenov et al., 2025). This criterion also exposes an activation-design question. In a modern decoder block, the nonlinear sublayer is a learned matrix–nonlinearity–matrix map. The activation sets local gain and curvature, while the adjacent matrices choose and recombine the coordinates on which that response acts. Standard training pipelines leave this local structure implicit to the optimizer. Activation coefficients and adjacent matrices are updated as ordinary parameters, leaving gain, coefficient-activity, and role information inside the sublayer unused by the matrix update.

We make this interface concrete with Rational Latent Basis (RLB) and MatrixPolicy. RLB maps residual states into grouped latent coordinates, normalizes each group by its RMS radius, and applies a trainable real-pole-free P5/Q4 rational response. The resulting response-gain, coefficient-activity, matrix-pressure, and pair-balance signals are detached from the forward graph and passed to MatrixPolicy. MatrixPolicy restricts their influence to the RLB input and output matrices; attention, embeddings, normalization parameters, rational coefficients, and ordinary Transformer tensors remain on AdamW. This local division exposes the quantities created by the nonlinear sublayer while preserving AdamW for the rest of the model.

We evaluate target arrival by measuring the tokens and wall-clock time required to reach a fixed validation-loss target under the same model, corpus, and protocol. In the completed study, the same 12-layer, width-768 causal Transformer is trained on DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4. The experiments include RLB-only controls and matched SiLU with AdamW and SiLU with Muon baselines under the same model, data, and measurement protocol. MatrixPolicy reaches shared validation-loss targets earlier at both 100M and 300M token budgets. [todo: Insert the larger-scale training result here.] The contribution is an activation design that exposes optimizer-visible matrix roles, a localized role-aware optimizer for the exposed matrix pair, and matched evidence that this interface reduces target-arrival tokens and wall-clock time across five pretraining corpora.

2 RELATED WORK

A Transformer nonlinear sublayer couples a scalar response to learned projections that form and recombine latent features. The original Transformer and T5 used prescribed ReLU responses, while GELU and Swish/SiLU supplied alternative fixed pointwise functions (Vaswani et al., 2017; Raffel et al., 2020; Hendrycks & Gimpel, 2016; Ramachandran et al., 2017). Gated linear units (GLUs) added a parallel projection and multiplicative gate, and SwiGLU adapted this topology to compute-matched Transformer sublayers (Dauphin et al., 2017; Shazeer, 2020). Gated nonlinear sublayers now appear across PaLM, LLaMA, Llama 3, Qwen2.5, DeepSeek-V3, Qwen3, and OLMo 2 (Chowdhery et al., 2022; Touvron et al., 2023; Grattafiori et al., 2024; Qwen et al., 2025; DeepSeek-AI et al., 2024; Yang et al., 2025; Team OLMo et al., 2025). Across these architectures, pretraining learns the projections around a response family selected in advance. Rational responses make that family trainable. Rational-network theory supports this function class through efficient approximation of smooth functions (Boullé et al., 2020) and expressivity separations from broad classes of fixed responses, including gated Transformer nonlinearities (Tang & Townsend, 2026). End-to-end coefficient learning has been demonstrated in Padé Activation Units and in Transformer pretraining and fine-tuning (Molina et al., 2020; Fang et al., 2023). Once learned between two projections, the response determines the realized feature in the output-projection gradient and the local Jacobian in the input-projection gradient. Response learning therefore changes both adjacent matrix gradients, as derived in Appendix C.1.

The optimizer then transforms these response-dependent matrix gradients. Its adaptation scope ranges from individual coordinates to full matrices. Coordinate-wise adaptation underlies Adam and AdamW, while memory-efficient variants factor or reorganize adaptive state in Adafactor and CAME (Kingma & Ba, 2015; Loshchilov & Hutter, 2019; Shazeer & Stern, 2018; Luo et al., 2023). At the same local scope, Lion, Schedule-Free methods, and cautious optimizers alter update signs, iterate averaging, or gradient-aligned masking (Chen et al., 2023; Defazio et al., 2024; Liang et al., 2026). Richer parameter structure enters through diagonal-curvature estimates in Sophia, block-shared adaptive rates in Adam-mini, and low-rank matrix-gradient subspaces in GaLore (Liu et al., 2024; Zhang et al., 2025; Zhao et al., 2024). Full-matrix methods extend this scope further. Shampoo preconditioned tensor axes, SOAP applies Adam in an evolving Shampoo eigenbasis, and Muon orthogonalizes matrix momentum (Gupta et al., 2018; Vyas et al., 2025; Liu et al., 2025). Across this progression, adaptation expands from coordinates to blocks, subspaces, tensor axes, and singular directions. Comparisons across these optimizer families span compute-time frontiers (Shah et al., 2025), fixed-token and GPU-hour budgets (Schlotthauer et al., 2025), and standardized pretraining scenarios (Semenov et al., 2025). Optimizer rankings also vary with hyperparameter tuning (Zhao et al., 2025; Wen et al., 2026) and with model or data scale (Wen et al., 2026; Volkova et al., 2026).

Within this literature, MuonClip is the closest signal-conditioned precedent because it uses the maximum attention logit to trigger post-update rescaling of query and key matrices (Kimi Team et al., 2025). Rational activation work exposes response-specific local structure because the response feature and Jacobian shape the two adjacent nonlinear-sublayer gradients, making response-conditioned updates to those paired projections the unresolved interface in the work reviewed here.

3 METHOD

RLB and MatrixPolicy are designed as a single interface. RLB defines the nonlinear sublayer and the detached summaries it exposes; MatrixPolicy consumes those summaries to update the two adjacent matrices. Rational coefficients and ordinary Transformer parameters remain on the AdamW path in the evaluated configuration.

Throughout the method we use

$$\text{clip}(x, a, b) = \min\{\max\{x, a\}, b\}, \quad (1)$$

$$\text{smoothstep}(a, b, x) = 3\omega^2 - 2\omega^3, \quad \omega = \text{clip}\left(\frac{x-a}{b-a}, 0, 1\right). \quad (2)$$

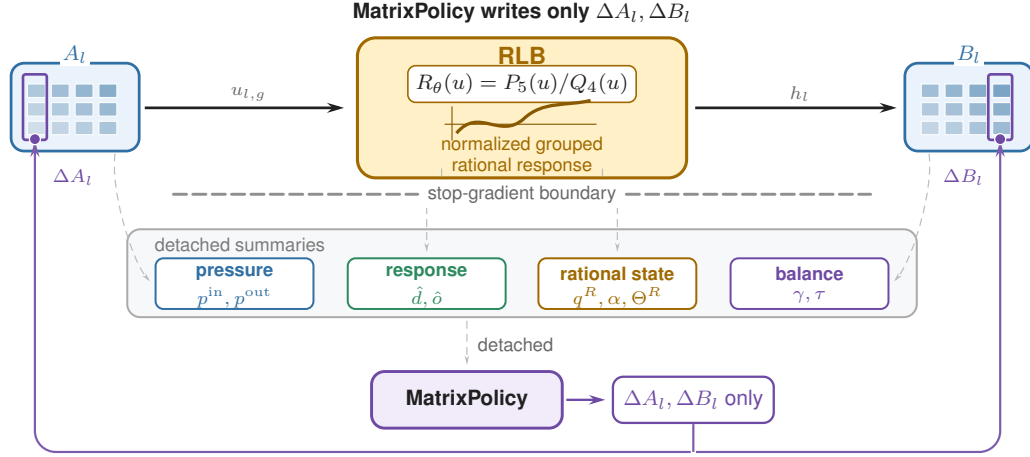


Figure 1: Activation-optimizer interface for RLB. In layer l , A_l maps the sublayer input into normalized rational groups and B_l recombines the rational features $h_l = \text{concat}_{g=1}^G h_{l,g}$. Forward RLB statistics, gradient pressure/activity EMAs, and pair-balance measurements cross a stop-gradient boundary as policy inputs. MatrixPolicy reads these summaries and writes ΔA_l and ΔB_l ; rational coefficients are AdamW-trained and policy-observed.

3.1 RATIONAL LATENT BASIS

For layer $l \in \{1, \dots, L\}$, let $x_l \in \mathbb{R}^d$ be the input to the nonlinear sublayer. RLB uses an input matrix $A_l \in \mathbb{R}^{H \times d}$ and an output matrix $B_l \in \mathbb{R}^{d \times H}$ with latent width $H = Gm$. Let $A_{l,g} \in \mathbb{R}^{m \times d}$ denote the row block of A_l , and let $B_{l,g} \in \mathbb{R}^{d \times m}$ denote the matching column block of B_l . In the implementation, A_l and B_l are the bias-free `in_proj.weight` and `out_proj.weight` of the nonlinear sublayer. The evaluated runs use an RMS floor $\epsilon_{\text{rms}} = 10^{-6}$. Each group has a trainable P5/Q4 rational response

$$P_{l,g}(u) = \sum_{i=0}^5 a_{l,g,i} u^i, \quad (3)$$

$$Q_{l,g}(u) = 1 + |b_{l,g,1}| |u| + |b_{l,g,2}| u^2 + |b_{l,g,3}| |u|^3 + |b_{l,g,4}| u^4, \quad (4)$$

$$R_{l,g}(u) = P_{l,g}(u)/Q_{l,g}(u). \quad (5)$$

Define the preactivation, RMS radius, normalized coordinate, and grouped rational feature by

$$z_{l,g}(x_l) = A_{l,g} x_l, \quad \rho_{l,g}(x_l) = \sqrt{m^{-1} \|z_{l,g}(x_l)\|_2^2 + \epsilon_{\text{rms}}}, \quad (6)$$

$$u_{l,g}(x_l) = \frac{z_{l,g}(x_l)}{\rho_{l,g}(x_l)}, \quad h_{l,g}(x_l) = \rho_{l,g}(x_l) R_{l,g}(u_{l,g}(x_l)), \quad h_l = \text{concat}_{g=1}^G h_{l,g}. \quad (7)$$

The RLB forward map is the single block expression

$$y_l = B_l h_l = \sum_{g=1}^G B_{l,g} \rho_{l,g}(x_l) R_{l,g} \left(\frac{A_{l,g} x_l}{\rho_{l,g}(x_l)} \right). \quad (8)$$

The scalar functions $P_{l,g}$, $Q_{l,g}$, and $R_{l,g}$ are applied coordinatewise when their input is a vector. Since $Q_{l,g}(u) \geq 1$ for all real u , the implemented response is pole-free on $\mathbb{R}$. The response is piecewise rational because of the absolute-value terms; derivative statistics use the autodiff convention $\text{sign}(0) = 0$ and equations involving $R'_{l,g}$ are interpreted almost everywhere. Numerator and denominator coefficients are initialized by a SiLU fit on $[-5, 5]$ and are then trained by AdamW. The evaluated model uses these groupwise P5/Q4 rational responses.

The forward map in equation 8 separates the normalized coordinate at which the rational response is evaluated from the group radius returned to the residual stream. This radius-factor form exposes

the positive homogeneity behind the matrix-pair coordinate formalized in Appendix B. The floor $\epsilon_{\text{rms}} > 0$ is an additive squared-RMS stabilizer. For fixed curves $R_l = \{R_{l,g}\}_{g=1}^G$, write this block map as $f_l(x; A_l, B_l, R_l)$. For any fixed input x such that $A_{l,g}x \neq 0$ for every group, if $\epsilon_{\text{rms}} = 0$, then for any $\lambda \in \mathbb{R}_{>0}^G$ and $D_l(\lambda) = \text{diag}(\lambda_1 I_m, \dots, \lambda_G I_m)$,

$$f_l(x; A_l, B_l, R_l) = f_l(x; D_l(\lambda)A_l, B_l D_l(\lambda)^{-1}, R_l). \quad (9)$$

This positive matrix-pair rescaling is exact in the unfloored map. The RMS floor perturbs the orbit, and MatrixPolicy follows the resulting approximate coordinate with bounded steps.

3.2 MATRIXPOLICY

MatrixPolicy is the optimizer-side policy attached to the RLB matrix pair (A_l, B_l) . We use the name for the matrix-specific controls in the evaluated optimizer configuration. Other Transformer tensors follow their assigned backbone optimizer, and the rational coefficients contribute activity signals while retaining their AdamW no-decay updates in the reported runs. The policy reads detached summaries. These summaries are produced by the RLB forward hook, by post-global-clipping gradients observed before MatrixPolicy rescales them, and by fixed-grid probes of the rational response. They feed, in order, group-gradient gating, a role/depth matrix step, and reciprocal pair rescaling.

This ordering matters. Let t be the one-based optimizer step after the wrapper increments its counter, T the planned number of steps, and $\xi_t = t/T$. The surrounding optimizer wrapper first constructs detached gradient memories from the current step; the inner matrix optimizer uses those memories to rescale matrix gradients and apply AdamW/Muon updates; the wrapper then measures the updated matrix pair and applies the reciprocal rescale. No summary tensor enters the forward graph. For notation, set $M_{l,\text{in}} = A_l$ and $M_{l,\text{out}} = B_l$, with $\sigma \in \{\text{in}, \text{out}\}$ denoting the two roles. For any tensor V , define

$$\text{rms}(V) = \sqrt{\text{mean}(V^2)}, \quad \text{rms}_\epsilon(V) = \sqrt{\text{mean}(V^2) + \epsilon}. \quad (10)$$

The evaluated implementation uses $\epsilon_{\text{live}} = 10^{-6}$ for cached forward gains and $\epsilon_p = \epsilon_{\text{probe}} = 10^{-8}$ for pressure, matrix-ratio, and fixed-grid probe floors.

The first family of summaries is the on-policy pressure and activity memory. Here “on-policy” means that the wrapper records statistics from the current training trajectory. All gradients in this paragraph are optimizer-visible loss gradients after global clipping and before MatrixPolicy group-gradient scaling. Let $\Theta_{l,g}^R$ collect the trainable rational-response coefficient blocks in group (l, g) ; for the reported P5/Q4 response these are the numerator and denominator coefficient tensors. The count $|\Theta_{l,g}^R|$ is a block count, and each mean is over scalar entries in one block. The instantaneous detached measurements are

$$p_{l,g}^{\text{in}} = \frac{\text{rms}_{\epsilon_p}(\nabla A_{l,g})}{\text{rms}_{\epsilon_p}(A_{l,g})}, \quad p_{l,g}^{\text{out}} = \frac{\text{rms}_{\epsilon_p}(\nabla B_{l,g})}{\text{rms}_{\epsilon_p}(B_{l,g})}, \quad (11)$$

$$p_{l,g}^R = \left[\frac{1}{|\Theta_{l,g}^R|} \sum_{\theta \in \Theta_{l,g}^R} \text{mean}((\nabla \theta)^2) + \epsilon_p \right]^{1/2}. \quad (12)$$

The matrix quantities are relative RMS-gradient proxies, while $p_{l,g}^R$ is raw coefficient-gradient energy. The wrapper stores detached EMA memories q^{in} , q^{out} , and q^R ,

$$q_{l,g,t}^r = 0.95 q_{l,g,t-1}^r + 0.05 p_{l,g,t}^r, \quad r \in \{\text{in}, \text{out}, R\}, \quad (13)$$

initialized from the first observed values. We reserve q in this subsection for these detached policy memories; the rational denominator remains denoted by $Q_{l,g}$. The two contrasts used by all three controls are

$$\pi_{l,g} = \log q_{l,g}^{\text{in}} - \log q_{l,g}^{\text{out}}, \quad \alpha_{l,g} = \log q_{l,g}^R - \frac{1}{2} (\log q_{l,g}^{\text{in}} + \log q_{l,g}^{\text{out}}). \quad (14)$$

The pressure contrast $\pi_{l,g}$ compares the two adjacent matrix roles, and the activity contrast $\alpha_{l,g}$ compares rational-coefficient gradient energy with the geometric mean of those role memories. Appendix C.1 derives the local gradient geometry behind these proxies. We drop the step subscript when the time is clear.

The second family of summaries comes from the RLB forward-statistics hook. The hook samples normalized coordinates and caches per-group response and derivative RMS values,

$$\hat{d}_{l,g}^{\text{live}} = \sqrt{\text{mean}(R'_{l,g}(u_{l,g})^2) + \epsilon_{\text{live}}}, \quad \hat{o}_{l,g}^{\text{live}} = \sqrt{\text{mean}(R_{l,g}(u_{l,g})^2) + \epsilon_{\text{live}}}. \quad (15)$$

The mean is over sampled scalar normalized coordinates for group (l, g) across the most recent refreshed sample, sequence positions, and the m group coordinates. The derivative gain is the input-role gain proxy, and the response gain is the output-role gain proxy. If a cache is unavailable at a step, the implementation sets the corresponding gain multiplier to identity while keeping the pressure/activity memories active.

The third family supplies the post-step pair rescale. Let $\mathcal{U} = \{u_k\}_{k=1}^{129}$ be a uniform probe grid over $[-5, 5]$, and define $\text{rms}_{\mathcal{U}}(\varphi) = \sqrt{|\mathcal{U}|^{-1} \sum_{u_k \in \mathcal{U}} \varphi(u_k)^2}$. The fixed-grid probe producer refreshes every ten optimizer steps and emits

$$\hat{d}_{l,g}^{\text{probe}} = \sqrt{\text{rms}_{\mathcal{U}}(R'_{l,g})^2 + \epsilon_{\text{probe}}}, \quad \hat{o}_{l,g}^{\text{probe}} = \sqrt{\text{rms}_{\mathcal{U}}(R_{l,g})^2 + \epsilon_{\text{probe}}}. \quad (16)$$

These probes describe rational curve shape on a shared interval; they are separate from the sampled live gains in equation 15. After the role/depth matrix step and immediately before reciprocal rescaling, the wrapper also measures the detached pair-state coordinate

$$\gamma_{l,g} = \log \text{rms}_{\epsilon_p}(A_{l,g}) - \log \text{rms}_{\epsilon_p}(B_{l,g}). \quad (17)$$

Only the probe gains and $\gamma_{l,g}$ are emitted by this summary family. The third control then computes $\tau_{l,g}$ and $\chi_{l,g}^{\text{rs}}$ from those summaries and from the pressure and activity memories.

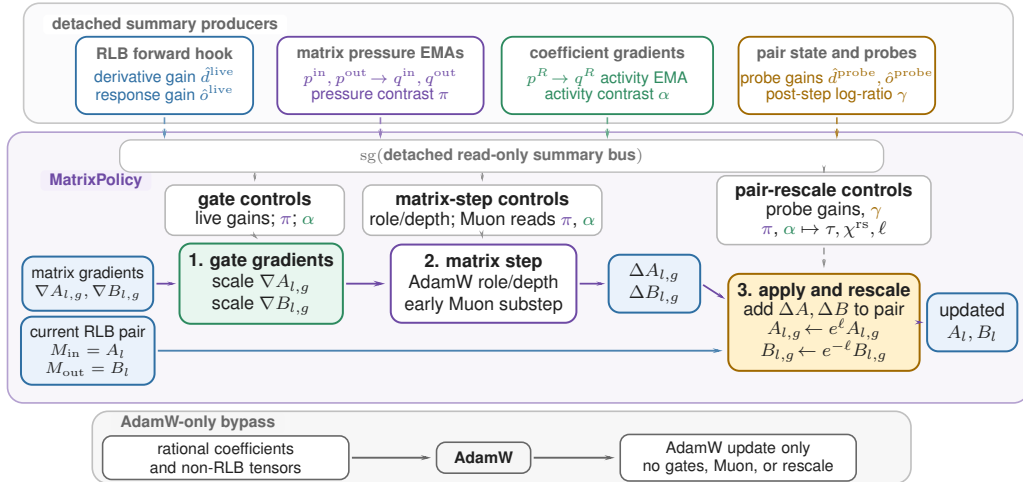


Figure 2: Detached-summary route for the evaluated MatrixPolicy system. The top band produces stop-gradient summaries from forward hooks, pre-policy gradients, fixed-grid probes, and post-step pair-state measurements; the read-only bus routes them to three MatrixPolicy consumers. The lower policy band separates the gradient/update lane from the current matrix-pair state lane, while rational coefficients and non-RLB tensors remain AdamW-only.

3.2.1 GROUP-GRADIENT GATING

The first control redistributes gradient scale among groups inside each matrix role. It reads the sampled live gains, the pressure contrast $\pi_{l,g}$, and the activity contrast $\alpha_{l,g}$, then produces scaled gradients for A_l and B_l . A smooth gate $\phi_t = \text{smoothstep}(0.02, 0.30, \xi_t)$ turns on this control in the reported runs. The evaluated manifest sets gain, pressure, and activity strengths to 0.20,

0.10, and 0.20, and clips group multipliers to $[0.75, 1.35]$. For positive group values v_j , define $\text{geomean}_{j=1}^G(v_j) = \exp\left(G^{-1} \sum_{j=1}^G \log v_j\right)$. The group-gradient rule is

$$\begin{aligned}
 \tilde{\pi}_{l,g} &= \text{clip}(\pi_{l,g}, -1.50, 1.50), & c_{l,g}^{\text{activity}} &= \exp\left[-0.20\phi_t \text{ReLU}\left(\frac{\alpha_{l,g} - 0.05}{0.45}\right)\right], \\
 c_{l,g,\text{in}}^{\text{gain}} &= \left(\frac{\text{geomean}_{j=1}^G \hat{d}_{l,j}^{\text{live}}}{\hat{d}_{l,g}^{\text{live}}}\right)^{0.20\phi_t}, & c_{l,g,\text{out}}^{\text{gain}} &= \left(\frac{\text{geomean}_{j=1}^G \hat{o}_{l,j}^{\text{live}}}{\hat{o}_{l,g}^{\text{live}}}\right)^{0.20\phi_t}, \\
 c_{l,g,\text{in}}^{\text{pressure}} &= \exp(-0.10\phi_t \tilde{\pi}_{l,g}), & c_{l,g,\text{out}}^{\text{pressure}} &= \exp(+0.10\phi_t \tilde{\pi}_{l,g}), \\
 c_{l,g,\sigma}^{\text{raw}} &= c_{l,g,\sigma}^{\text{gain}} c_{l,g,\sigma}^{\text{pressure}} c_{l,g}^{\text{activity}}, & c_{l,g,\sigma} &= \text{clip}\left(\frac{c_{l,g,\sigma}^{\text{raw}}}{\text{geomean}_{j=1}^G c_{l,j,\sigma}^{\text{raw}}}, 0.75, 1.35\right).
 \end{aligned} \tag{18}$$

Live derivative gain controls the input role, live response gain controls the output role, and pressure acts with opposite signs on the two roles. When $\pi_{l,g} > 0$, the input-side relative-gradient EMA is larger than the output-side EMA, so the gate reduces the input-role gradient scale and increases the output-role gradient scale for that group. The activity term damps groups whose rational coefficient gradients are large relative to the adjacent matrix memories. The geometric recentering preserves the role-wise mean log multiplier before clipping; Appendix C.2 gives the log-space identity.

The scaled matrix gradients are

$$\tilde{\nabla} A_{l,g} = c_{l,g,\text{in}} \nabla A_{l,g}, \quad \tilde{\nabla} B_{l,g} = c_{l,g,\text{out}} \nabla B_{l,g}. \tag{19}$$

Let $\tilde{\nabla} M_{l,\text{in}}$ be the row-block concatenation of $\{\tilde{\nabla} A_{l,g}\}_{g=1}^G$, and let $\tilde{\nabla} M_{l,\text{out}}$ be the column-block concatenation of $\{\tilde{\nabla} B_{l,g}\}_{g=1}^G$.

3.2.2 ROLE/DEPTH MATRIX STEP

The second control applies the matrix optimizer step to the scaled gradients. It assigns different depth trends to the input and output roles, runs AdamW on both matrices, and activates a transient Muon substep early in training. In the reported configuration, the AdamW multiplier depends on role and depth; the pressure and activity memories enter this control through the Muon attenuation factor. Let $\delta_l = (l - 1)/(L - 1)$ for $L > 1$ and $\delta_l = 1/2$ otherwise. The branch schedules are

$$\begin{aligned}
 \varrho_{\text{in}}(l) &= \text{clip}(1 - 0.50(\delta_l - 0.5), 0.55, 1.40), \\
 \varrho_{\text{out}}(l) &= \text{clip}(1 + 1.00(\delta_l - 0.5), 0.55, 1.40), \\
 s_{l,\sigma}^{\text{Adam}} &= \text{clip}(3.0 \max\{0.10, 1 + 1.20(\varrho_{\sigma}(l) - 1)\}, 0.40, 4.0), \\
 \Psi_{l,t} &= \text{clip}\left(G^{-1} \sum_{g=1}^G |\pi_{l,g}|, 0, 1.50\right), \\
 \Omega_{l,t} &= \text{ReLU}\left(\frac{G^{-1} \sum_{g=1}^G \alpha_{l,g} - 0.05}{0.45}\right), \\
 \psi_{l,t} &= \text{clip}(\exp[-0.30\Psi_{l,t} - 0.65\Omega_{l,t}], 0.10, 1.15), \\
 \omega_t^\mu &= \text{smoothstep}(0.02, 0.12, \xi_t)[1 - \text{smoothstep}(0.20, 0.36, \xi_t)], \\
 \mu_{l,\sigma,t} &= \text{clip}(0.75 \omega_t^\mu \varrho_{\sigma}(l) \psi_{l,t}, 0, 0.75).
 \end{aligned} \tag{20}$$

Here $s_{l,\sigma}^{\text{Adam}}$ is the role/depth AdamW multiplier. The aggregate pressure $\Psi_{l,t}$ and activity $\Omega_{l,t}$ form the Muon attenuation $\psi_{l,t}$, so the active Muon fraction $\mu_{l,\sigma,t}$ is largest during the early Muon window and shrinks when the detached summaries indicate large role imbalance or large rational-coefficient activity. For $M_{l,\text{in}} = A_l$ and $M_{l,\text{out}} = B_l$, let η_t be the base learning rate and let $s_\mu = 1.0$ be the evaluated Muon learning-rate scale. MatrixPolicy applies an AdamW matrix step followed by any active Muon substep. The gate $\mu_{l,\sigma,t}$ scales the two optimizer learning rates; AdamW and Muon

keep separate states and are applied sequentially:

$$\begin{aligned}\widetilde{M}_{l,\sigma}^{t+1} &= \text{AdamW}_{\eta_t s_{l,\sigma}^{\text{Adam}}(1-\mu_{l,\sigma,t})}(M_{l,\sigma}^t; \widetilde{\nabla} M_{l,\sigma}^t), \\ M_{l,\sigma}^{t+1} &= \text{Muon}_{\eta_t s_{\mu} \mu_{l,\sigma,t}}(\widetilde{M}_{l,\sigma}^{t+1}; \widetilde{\nabla} M_{l,\sigma}^t),\end{aligned}\tag{21}$$

The AdamW step uses the training-protocol betas, weight decay, and learning-rate schedule in Appendix A.1. The Muon substep is the `torch.optim.Muon` update on two-dimensional RLB matrices with momentum 0.95, five Newton–Schulz steps, and match-RMS learning-rate adjustment (Liu et al., 2025); it becomes inactive when all $\mu_{l,\sigma,t}$ decay to zero. Appendix C.3 states the matrix-spectrum geometry behind this transient branch.

3.2.3 RECIPROCAL PAIR RESCALING

The third control chooses a bounded positive representative of the same RLB matrix pair after the matrix update. It is implemented by the surrounding transport wrapper and is evaluated every five optimizer steps after the AdamW/Muon branch; fixed-grid probe gains are refreshed every ten optimizer steps and reused between refreshes. It consumes the probe gains, the post-step log ratio $\gamma_{l,g}$, and the pressure/activity memories. The target combines the probe-gain balance coordinate derived in Appendix C.4 with a bounded pressure offset:

$$\begin{aligned}\bar{\pi}_{l,g} &= \text{clip}(\pi_{l,g}, -1.25, 1.25), \\ \tau_{l,g} &= \frac{1}{2} \left(\log \hat{d}_{l,g}^{\text{probe}} - \log \hat{o}_{l,g}^{\text{probe}} \right) + 0.25 \bar{\pi}_{l,g}, \\ \bar{\alpha}_{l,g} &= \text{clip}(\alpha_{l,g}, -1.25, 1.25), \\ \chi_{l,g}^{\text{rs}} &= \exp(0.10 \bar{\alpha}_{l,g}), \quad (\text{rescale-speed factor}) \\ s_{l,t} &= 0.50 \text{ smoothstep}(0.03, 0.35, \xi_t) \text{ clip}(1 + 0.15(\delta_l - 0.5), 0.75, 1.25).\end{aligned}\tag{22}$$

The current log ratio $\gamma_{l,g}$ is measured after the matrix step. The clipped log step and reciprocal update are

$$\begin{aligned}\ell_{l,g,t} &= \text{clip}\left(\frac{1}{2}[\tau_{l,g} - \gamma_{l,g}]s_{l,t}\chi_{l,g}^{\text{rs}}, -0.030, 0.030\right), \\ A_{l,g} &\leftarrow e^{\ell_{l,g,t}} A_{l,g}, \quad B_{l,g} \leftarrow e^{-\ell_{l,g,t}} B_{l,g}.\end{aligned}\tag{23}$$

The first term of $\tau_{l,g}$ equalizes derivative and response probe gains in the positive log coordinate, the pressure term shifts the target within a bounded range, and χ^{rs} changes the clipped rescale speed. Larger $\alpha_{l,g}$ accelerates this representative-balancing move, while the earlier gradient gate and Muon attenuation use the same activity contrast to reduce gradient-driven matrix motion. In the nonzero, unfloored idealization of equation 9, the reciprocal pair move preserves the represented function; near the RMS floor it is a clipped approximate balance step.

4 EXPERIMENTS

4.1 FIXED-SCALE TARGET-ARRIVAL STUDY

The fixed-scale study uses the same 12-layer, width-768 causal Transformer on five corpora: DCLM/DataComp-LM (Li et al., 2024), FineWeb-Edu and FineWeb (Penedo et al., 2024), Dolma-sample (Soldaini et al., 2024), and C4 (Raffel et al., 2020). The 100M-token budget uses 3050 optimizer steps and the 300M-token budget uses 9150 optimizer steps, with three seeds per method/dataset pair. The main comparison keeps the architecture and data fixed while changing the activation and optimizer used in the nonlinear sublayer: MatrixPolicy is compared with SiLU controls and with RLB-only controls, where RLB-only places the same activation under the named standard optimizer. Shared optimizer, evaluation-cadence, and throughput-measurement details are reported in Appendix A.1; broader optimizer comparisons beyond AdamW and Muon are summarized in Table 2.

Table 1: Target-arrival efficiency for the fixed 12-layer, width-768 Transformer. RLB denotes the same rational latent-basis activation without MatrixPolicy. At the hardest validation-loss target reached by every listed method in all three seeds, MatrixPolicy is always fastest; each token-saving column reports MatrixPolicy’s savings relative to the method named below it.

Budget	Dataset	Target loss	MatrixPolicy token savings (%)				Time to target (min)				
			vs. SiLU AdamW	vs. SiLU Muon	vs. RLB AdamW	vs. RLB Muon	Matrix Policy	SiLU AdamW	SiLU Muon	RLB AdamW	RLB Muon
100M	DCLM	4.55	20.7	32.4	20.7	34.3	12.8	20.8	26.0	14.3	19.0
	FineWeb-Edu	4.40	19.5	29.5	20.2	29.5	12.9	20.3	24.4	14.2	18.0
	FineWeb	4.60	22.8	32.1	21.5	32.1	12.6	16.4	20.2	16.2	20.1
	Dolma	4.60	22.0	34.0	22.7	34.9	12.8	17.6	22.0	17.3	22.2
	C4	4.60	20.7	29.2	19.3	28.7	11.6	14.3	17.3	14.9	18.8
300M	DCLM	4.10	23.3	8.3	24.0	7.0	39.6	53.5	45.9	54.7	45.6
	FineWeb-Edu	3.85	22.8	6.7	22.6	4.8	40.0	59.9	47.9	52.3	50.0
	FineWeb	4.10	23.1	7.1	22.4	5.3	40.8	61.2	46.4	55.6	46.6
	Dolma	3.95	22.3	9.8	22.3	6.5	44.4	49.3	52.3	50.5	47.4
	C4	4.00	26.4	8.5	25.3	6.5	41.8	64.7	55.5	59.4	51.9

Table 1 is the primary empirical result. It asks: for the hardest common validation-loss target reached by every listed method in all three seeds, what fraction of target-arrival tokens does MatrixPolicy save, and how many measured target-arrival minutes does each method require? Across all five datasets under the 100M-token budget, MatrixPolicy saves 19.5–22.8% of the SiLU with AdamW target-arrival tokens and 29.2–34.0% of the SiLU with Muon target-arrival tokens. Under the 300M-token budget, after longer training compresses endpoint gaps, MatrixPolicy still saves 22.3–26.4% versus SiLU with AdamW and 6.7–9.8% versus SiLU with Muon, with lower target-arrival time in every listed comparison.

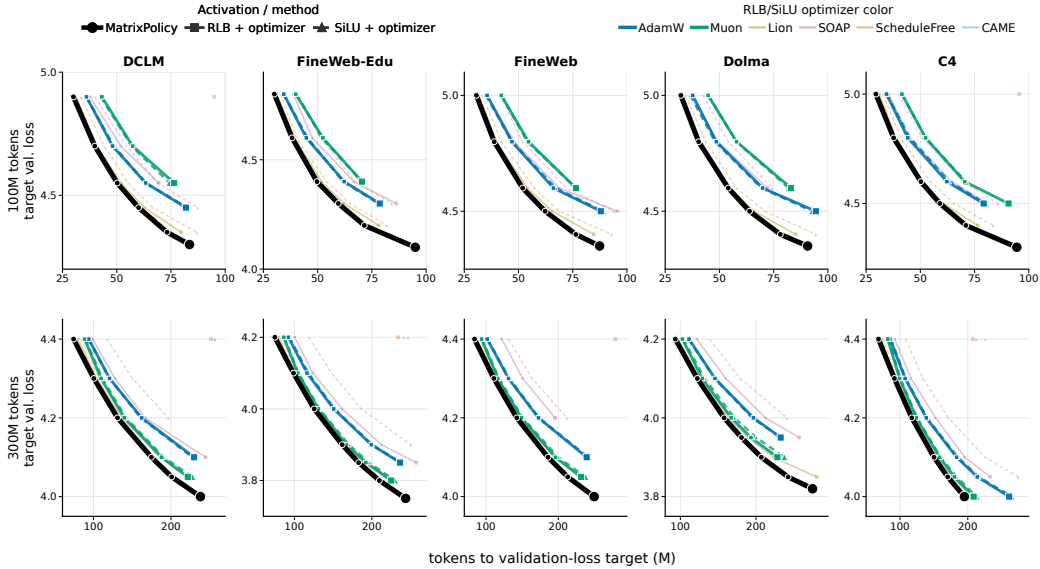


Figure 3: MatrixPolicy shifts target arrival left across the completed fixed-scale study. Rows show the 100M- and 300M-token budgets, and columns show datasets. Each point is the mean token count needed to reach one validation-loss target in all three seeds; connected points trace one method from easier to harder targets. Black circles are MatrixPolicy. For every optimizer color, solid squares are RLB with that optimizer and dashed triangles are SiLU with the same optimizer. Lower targets are harder, and the hard-target timing and token-saving percentages are quantified in Table 1.

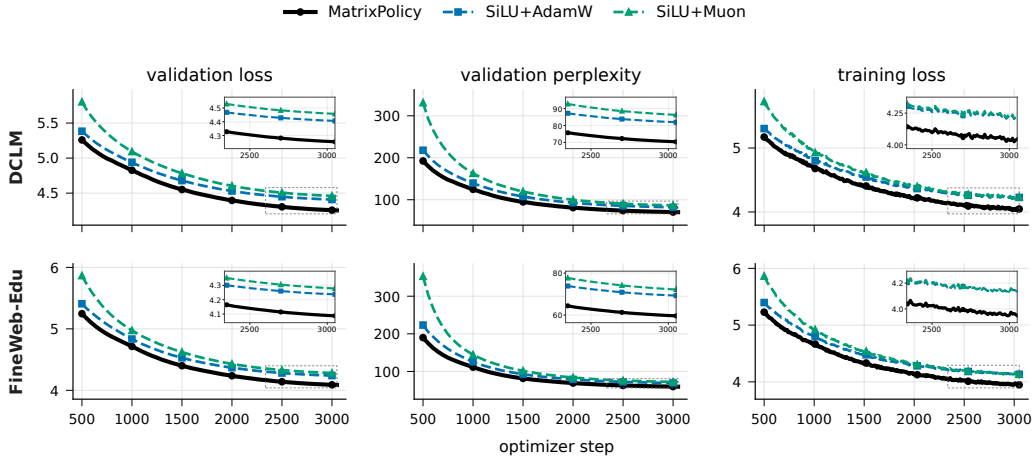


Figure 4: Representative 100M-token-budget training dynamics on DCLM and FineWeb-Edu. Rows show datasets and columns show validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. This figure uses the two standard SiLU baselines so that the early target-arrival effect in Table 1 remains visually legible; inset windows zoom the final-quarter saturation region.

The 100M-token-budget curves show why target arrival is the right main readout. MatrixPolicy separates early in validation loss and validation perplexity while keeping training loss competitive. A long fixed budget eventually compresses many activation/optimizer endpoints, especially under the 300M-token budget, so the paper emphasizes the number of tokens and minutes needed to reach the same useful validation-loss levels.

4.2 TODO: LARGER-SCALE TRAINING RESULTS

TODO: Insert the larger-scale training result here.

4.3 TODO: COMPONENT ABLATIONS

TODO: Insert the component ablation study here.

5 CONCLUSION

RLB turns the nonlinear Transformer sublayer into an optimizer-visible interface by exposing grouped normalized coordinates, trainable groupwise rational responses, and an idealized positive matrix-pair relation used as a bounded balancing coordinate. MatrixPolicy uses that interface to update the RLB input and output matrices with role-aware statistics, while rational coefficients and non-RLB-matrix parameters use AdamW. The completed 100M-token and 300M-token runs show that MatrixPolicy reaches common validation losses with fewer tokens and lower measured target-arrival time than the listed SiLU and RLB controls. [todo: Complete this conclusion after the larger-scale and ablation results are added.]

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A EXPERIMENT APPENDIX

This appendix gives the fixed-scale protocol behind Section 4.1, the full training curves for the five corpora, and the broader optimizer endpoint comparison used to contextualize the main SiLU and RLB controls.

A.1 FIXED-SCALE TARGET-ARRIVAL STUDY

A.1.1 MODEL AND DATA PIPELINE

The fixed-scale experiments use a custom causal decoder Transformer. The block uses a LLaMA-style pre-normalized layout with RMSNorm before attention and before the nonlinear sublayer, bias-free query/key/value and output attention projections, rotary position embeddings, causal scaled-dot-product attention, residual attention and nonlinear-sublayer updates, a final RMSNorm, and a tied token-embedding/output head (Zhang & Sennrich, 2019; Su et al., 2021; Touvron et al., 2023). The fixed model has 12 layers, width 768, 12 heads of dimension 64, and context length 256.

The SiLU controls use a SwiGLU-like gated nonlinear sublayer with bias-free matrices $W_g, W_v \in \mathbb{R}^{768 \times 2048}$ and $W_o \in \mathbb{R}^{2048 \times 768}$, computing $(\text{SiLU}(xW_g) \odot xW_v)W_o$. MatrixPolicy and RLB-only configurations use a matched single-branch matrix pair. An input matrix maps width 768 to 3072 latent coordinates, the coordinates are partitioned into groups and transformed by the real-pole-free P5/Q4 rational response, and an output matrix maps the 3072 coordinates back to width 768. With the default group size 256, this gives $G = 12$ groups of width $m = 256$; the RMS floor is $\epsilon_{\text{rms}} = 10^{-6}$. This is the RLB configuration used by the main MatrixPolicy and RLB-control results.

Tokenization uses the GPT-2 tokenizer through the Hugging Face tokenizer interface with its base vocabulary; an EOS or pad token is appended between documents. Training samples random contiguous blocks of length 257 and forms next-token prediction pairs of length 256. The DCLM, FineWeb-Edu, FineWeb, Dolma-sample, and C4 runs use the text field specified in the manifests, with validation slices drawn from held-out token offsets of the same corpus stream.

A.1.2 TRAINING PROTOCOL

All runs use batch size 16, gradient accumulation 2, and therefore 32768 global tokens per optimizer step. The 100M-token budget consists of 3050 optimizer steps; the 300M-token budget consists of 9150 optimizer steps. Each method/dataset pair is run with seeds 1337, 2027, and 3407. Validation is performed every 50 optimizer steps, so token-to-target arrivals are quantized in increments of 1.64M tokens.

The MatrixPolicy rows in the main table use the RLB activation, whose active nonlinearity is the groupwise P5/Q4 response in equations 3–5. The evaluated configuration uses AdamW as the backbone optimizer and disables the separate rational-coefficient optimizer. It enables MatrixPolicy group redistribution with gain/pressure/activity strengths 0.20/0.10/0.20, schedule window 0.02–0.30, and group multiplier clip $[0.75, 1.35]$. The RLB matrix pair then uses the role/depth-scaled AdamW branch and the early Muon substep from Section 3.2. The bounded RLB pair-rescale wrapper runs every five optimizer steps after the matrix branch. Rational coefficients, attention weights, embeddings, normalization parameters, and ordinary Transformer weights use AdamW in this configuration.

Target-arrival minutes in Table 1 are computed seed-wise from the first validation event that reaches the shared target and the same run’s measured seconds per optimizer step. The per-step time is the training-script wall time recorded in the run summary divided by completed optimizer steps for that run; target arrivals are therefore measured with the same evaluation cadence and timing source across methods.

A.1.3 SUPPORTING FIXED-SCALE CURVES

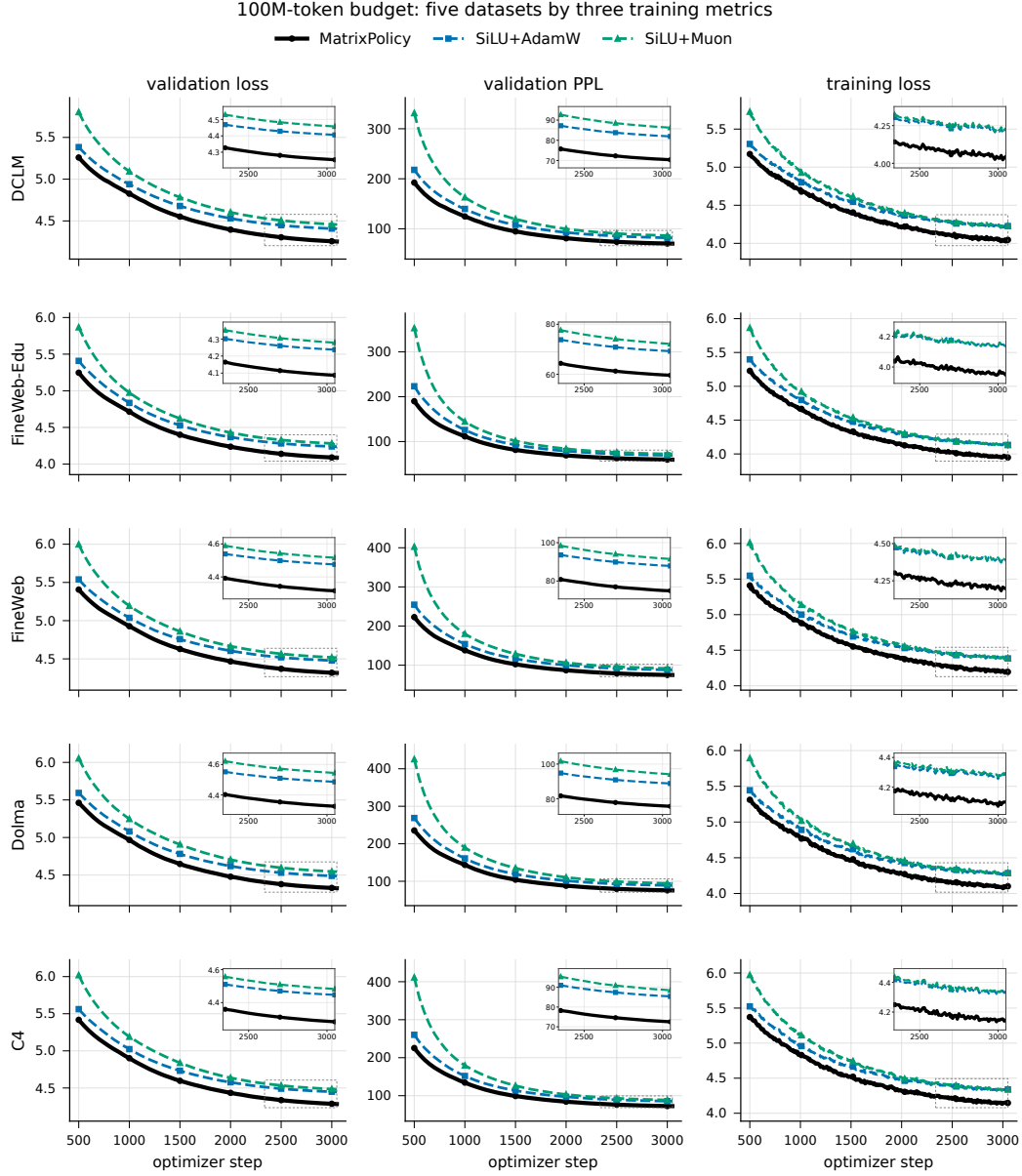


Figure 5: 100M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. Insets zoom the final-quarter saturation region in each panel.

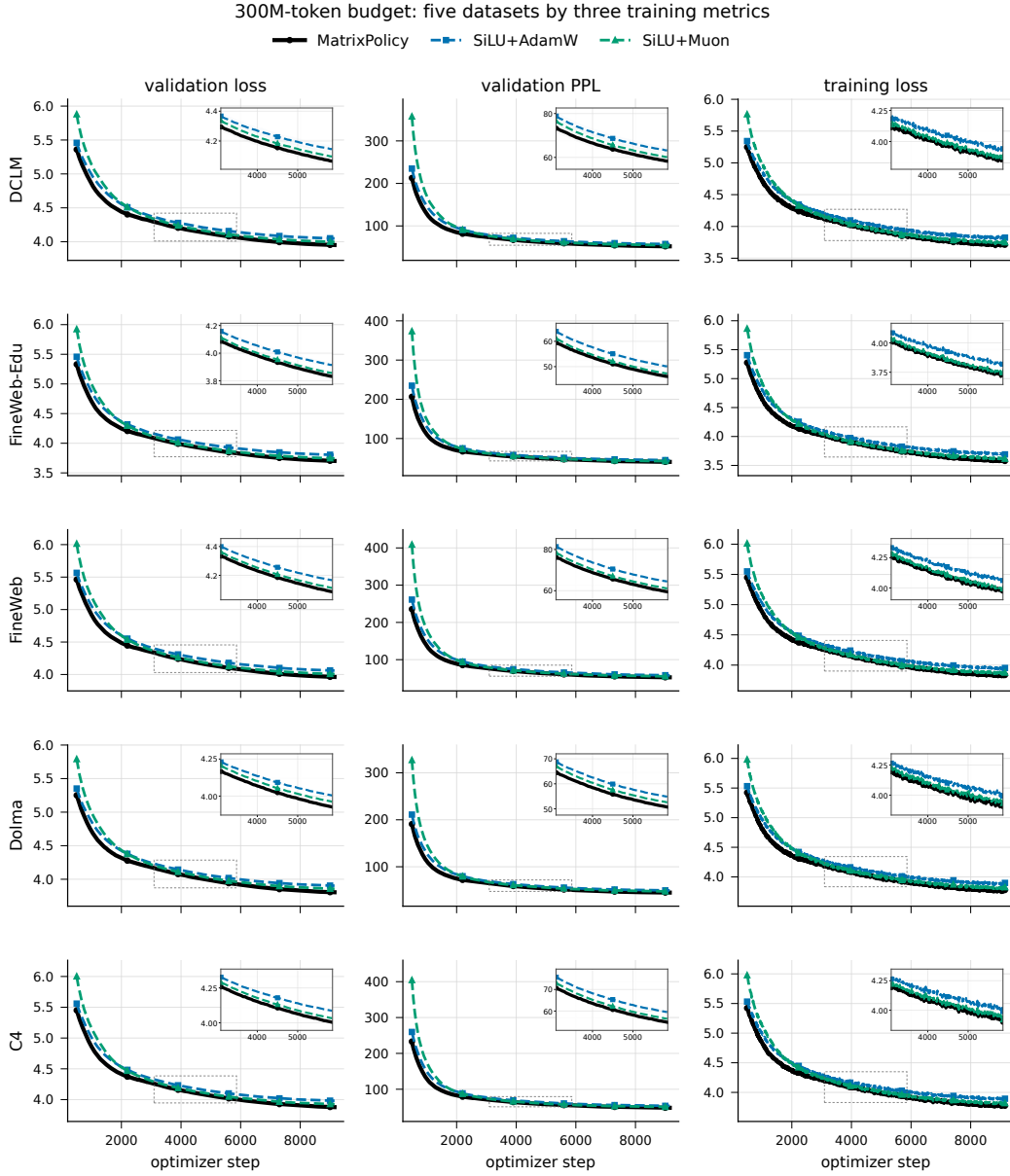


Figure 6: 300M-token-budget dynamics across all five datasets. Rows are datasets and columns are validation loss, validation perplexity, and training loss. Lines are seed means over three runs and bands show one sample standard deviation. Insets zoom the middle training window where early saturation differences are most visible.

A.1.4 BROAD OPTIMIZER ENDPOINT TABLE

The fixed-scale package also includes the broader optimizer endpoints referenced from Section 4.1. These rows use the same model, datasets, seeds, token budgets, and validation schedule as the main comparison, while replacing the nonlinear-sublayer optimizer according to the table entry.

Table 2: Final validation loss for the broader optimizer sweep on the fixed 12-layer, width-768 language model. Values are mean $\pm$ sample standard deviation over three seeds; lower is better. RLB rows use the same rational latent-basis activation without MatrixPolicy.

Budget	Method	DCLM	FineWeb-Edu	FineWeb	Dolma	C4
100M tokens	MatrixPolicy	4.254 $\pm$ 0.006	4.087 $\pm$ 0.010	4.316 $\pm$ 0.013	4.325 $\pm$ 0.005	4.284 $\pm$ 0.020
	SiLU + AdamW	4.406 $\pm$ 0.010	4.237 $\pm$ 0.009	4.476 $\pm$ 0.010	4.486 $\pm$ 0.001	4.447 $\pm$ 0.016
	SiLU + Lion	4.318 $\pm$ 0.007	4.149 $\pm$ 0.009	4.383 $\pm$ 0.008	4.388 $\pm$ 0.005	4.354 $\pm$ 0.016
	SiLU + Muon	4.457 $\pm$ 0.013	4.279 $\pm$ 0.024	4.516 $\pm$ 0.026	4.544 $\pm$ 0.011	4.482 $\pm$ 0.023
	SiLU + SOAP	4.416 $\pm$ 0.004	4.263 $\pm$ 0.022	4.484 $\pm$ 0.011	4.499 $\pm$ 0.004	4.458 $\pm$ 0.015
	SiLU + ScheduleFree	4.902 $\pm$ 0.011	4.826 $\pm$ 0.007	5.014 $\pm$ 0.019	5.049 $\pm$ 0.015	5.006 $\pm$ 0.016
	SiLU + CAME	5.011 $\pm$ 0.014	4.921 $\pm$ 0.012	5.133 $\pm$ 0.013	5.165 $\pm$ 0.019	5.123 $\pm$ 0.018
	RLB + AdamW	4.405 $\pm$ 0.008	4.239 $\pm$ 0.008	4.472 $\pm$ 0.014	4.488 $\pm$ 0.003	4.441 $\pm$ 0.016
	RLB + Lion	4.295 $\pm$ 0.008	4.136 $\pm$ 0.008	4.363 $\pm$ 0.011	4.362 $\pm$ 0.007	4.327 $\pm$ 0.016
	RLB + Muon	4.467 $\pm$ 0.012	4.283 $\pm$ 0.028	4.516 $\pm$ 0.006	4.548 $\pm$ 0.015	4.476 $\pm$ 0.010
	RLB + SOAP	4.436 $\pm$ 0.044	4.275 $\pm$ 0.024	4.689 $\pm$ 0.336	4.518 $\pm$ 0.036	4.446 $\pm$ 0.019
	RLB + ScheduleFree	4.887 $\pm$ 0.005	4.796 $\pm$ 0.013	4.999 $\pm$ 0.019	5.036 $\pm$ 0.013	4.984 $\pm$ 0.016
	RLB + CAME	5.014 $\pm$ 0.009	4.915 $\pm$ 0.006	5.134 $\pm$ 0.016	5.151 $\pm$ 0.017	5.111 $\pm$ 0.016
300M tokens	MatrixPolicy	3.952 $\pm$ 0.028	3.702 $\pm$ 0.021	3.962 $\pm$ 0.008	3.806 $\pm$ 0.007	3.878 $\pm$ 0.014
	SiLU + AdamW	4.049 $\pm$ 0.027	3.803 $\pm$ 0.018	4.061 $\pm$ 0.010	3.904 $\pm$ 0.009	3.981 $\pm$ 0.013
	SiLU + Lion	3.993 $\pm$ 0.023	3.744 $\pm$ 0.021	4.001 $\pm$ 0.008	3.848 $\pm$ 0.009	3.921 $\pm$ 0.011
	SiLU + Muon	3.997 $\pm$ 0.030	3.745 $\pm$ 0.017	4.007 $\pm$ 0.013	3.858 $\pm$ 0.010	3.925 $\pm$ 0.013
	SiLU + SOAP	4.096 $\pm$ 0.030	3.863 $\pm$ 0.020	4.114 $\pm$ 0.010	3.957 $\pm$ 0.009	4.035 $\pm$ 0.011
	SiLU + ScheduleFree	4.366 $\pm$ 0.030	4.156 $\pm$ 0.024	4.398 $\pm$ 0.011	4.215 $\pm$ 0.005	4.316 $\pm$ 0.011
	SiLU + CAME	4.368 $\pm$ 0.023	4.150 $\pm$ 0.021	4.406 $\pm$ 0.019	4.249 $\pm$ 0.027	4.330 $\pm$ 0.015
	RLB + AdamW	4.050 $\pm$ 0.030	3.802 $\pm$ 0.021	4.060 $\pm$ 0.011	3.905 $\pm$ 0.008	3.979 $\pm$ 0.010
	RLB + Lion	3.989 $\pm$ 0.029	3.742 $\pm$ 0.021	3.996 $\pm$ 0.011	3.841 $\pm$ 0.008	3.913 $\pm$ 0.014
	RLB + Muon	3.991 $\pm$ 0.026	3.737 $\pm$ 0.019	3.999 $\pm$ 0.011	3.848 $\pm$ 0.005	3.919 $\pm$ 0.015
	RLB + SOAP	4.061 $\pm$ 0.033	3.824 $\pm$ 0.013	4.108 $\pm$ 0.032	3.927 $\pm$ 0.013	4.002 $\pm$ 0.019
	RLB + ScheduleFree	4.361 $\pm$ 0.034	4.142 $\pm$ 0.022	4.386 $\pm$ 0.008	4.212 $\pm$ 0.011	4.308 $\pm$ 0.007
	RLB + CAME	4.450 $\pm$ 0.043	4.225 $\pm$ 0.036	4.480 $\pm$ 0.006	4.267 $\pm$ 0.041	4.365 $\pm$ 0.058

A.1.5 LEARNING-RATE AND WEIGHT-DECAY SENSITIVITY

We ran a 100M-token, single-seed sensitivity grid for MatrixPolicy and the two standard SiLU controls. The grid crosses learning rates $\{1, 2, 3, 5\} \times 10^{-4}$ with weight decays 0, 0.05, 0.10, and 0.20 on all five corpora, using the model, tokenizer, evaluation cadence, and token budget in Appendix A.1.2. Let d index a corpus, η denote the outer learning rate, λ the weight-decay coefficient, $h = (\eta, \lambda)$ their matched grid cell, and $L_{d,h,a}(s_k)$ be method a 's validation loss at $s_k \in \mathcal{S} = \{50, 100, \dots, 3050\}$. Writing MP for MatrixPolicy and c for either SiLU control, define the observed-checkpoint margin

$$m_{d,h,c} = \min_{s_k \in \mathcal{S}} [L_{d,h,c}(s_k) - L_{d,h,MP}(s_k)]. \quad (24)$$

A positive margin certifies that MatrixPolicy has lower validation loss at every observed checkpoint from step 50 onward in that matched cell. For a target ℓ , define its sampled first-hit token count as

$$N_{d,h,a}(\ell) = 32,768 \min\{s_k \in \mathcal{S} : L_{d,h,a}(s_k) \leq \ell\}, \quad (25)$$

with $N_{d,h,a}(\ell) = +\infty$ when the set is empty. Therefore $m_{d,h,c} > 0$ implies $N_{d,h,MP}(\ell) \leq N_{d,h,c}(\ell)$ whenever the control's first hit is finite.

Figure 7 gives strict observed-checkpoint dominance in all 80 dataset-grid cells against each control. Every one of the 4,880 paired checkpoint comparisons per control is positive; the smallest observed margin is 0.041 against SiLU with AdamW and 0.045 against SiLU with Muon.

Table 3 makes this target-arrival consequence explicit at the main 100M-token targets from Table 1 and at targets 0.10 and 0.20 loss units lower. Each row partitions all 16 cells and reports mutual non-arrivals as a separate outcome.

At the main targets, MatrixPolicy arrives earlier in all 40 mutual-arrival comparisons per control and is the only method to arrive in 20 additional cells; no cell favors a control. The same ordering holds at both harder target levels. Pooled over mutual arrivals, median token savings at the main, -0.10 , and -0.20 target levels are 15.7%, 19.3%, and 14.3% against SiLU with AdamW and 19.9%, 13.5%, and 10.3% against SiLU with Muon. Node assignments were not matched across the sensitivity sweep, so this analysis reports token arrivals only. The primary-study wall-clock comparison remains in Table 1. Together, the checkpoint and target analyses show the same convergence advantage at all 16 evaluated learning-rate and weight-decay settings for seed 1337.

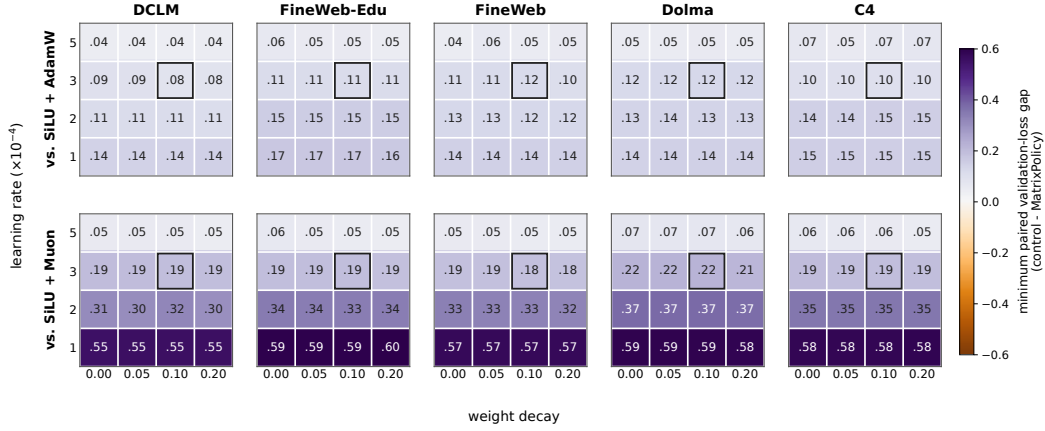


Figure 7: Paired observed-checkpoint margins over the complete learning-rate/weight-decay grid. Each tile is $m_{d,h,c}$ from equation 24 for one dataset, grid cell, and comparator at seed 1337. Positive values mean that MatrixPolicy has lower validation loss at every observed checkpoint from steps 50 through 3050. The dark outline marks the fixed main-study setting (3×10^{-4} , 0.10); all 160 tiles are positive.

Table 3: Complete-grid target arrival for seed 1337 at 100M tokens. Within each dataset, rows give the main target followed by targets 0.10 and 0.20 lower. Each comparator block partitions all 16 cells. “Both reach” lists MatrixPolicy earlier / control earlier / tied, and “only reaches” lists MatrixPolicy / control. Token savings are medians [minimum, maximum] of $100(N_c - N_{MP})/N_c$ over mutual arrivals, where N is first-hit tokens.

Dataset	Target loss	Compared with SiLU + AdamW				Compared with SiLU + Muon			
		Both reach MP / control / tie	Only reaches MP / control	Neither reaches	Token saved (%) median [range]	Both reach MP / control / tie	Only reaches MP / control	Neither reaches	Token saved (%) median [range]
DCLM	4.55	8/0/0	4/0	4	14.4 [7.1,18.4]	8/0/0	4/0	4	19.1 [10.3,29.5]
	4.45	8/0/0	4/0	4	16.9 [8.8,26.5]	7/0/0	5/0	4	11.4 [8.8,40.0]
	4.35	4/0/0	4/0	8	11.9 [11.6,14.0]	4/0/0	4/0	8	10.8 [9.5,11.9]
FineWeb-Edu	4.40	8/0/0	4/0	4	14.4 [10.3,18.4]	8/0/0	4/0	4	19.1 [10.3,27.9]
	4.30	8/0/0	4/0	4	19.6 [11.4,26.5]	8/0/0	4/0	4	23.2 [8.8,37.9]
	4.20	4/0/0	4/0	8	14.0 [14.0,16.3]	4/0/0	4/0	8	9.8 [7.5,12.2]
FineWeb	4.60	8/0/0	4/0	4	16.4 [10.3,22.5]	8/0/0	4/0	4	19.1 [10.3,29.5]
	4.50	8/0/0	4/0	4	21.4 [11.4,28.8]	8/0/0	4/0	4	22.0 [8.8,38.3]
	4.40	4/0/0	4/0	8	15.9 [13.6,15.9]	4/0/0	4/0	8	10.8 [9.5,11.9]
Dolma	4.60	8/0/0	4/0	4	18.1 [10.0,23.3]	8/0/0	4/0	4	22.1 [12.9,34.0]
	4.50	8/0/0	4/0	4	22.5 [13.5,32.8]	4/0/0	8/0	4	13.5 [11.1,13.5]
	4.40	4/0/0	4/0	8	14.3 [13.3,15.2]	4/0/0	4/0	8	13.3 [11.4,13.3]
C4	4.60	8/0/0	4/0	4	14.8 [10.7,18.9]	8/0/0	4/0	4	18.8 [10.7,28.6]
	4.50	8/0/0	4/0	4	19.3 [14.7,25.5]	8/0/0	4/0	4	21.7 [9.4,34.0]
	4.40	4/0/0	5/0	7	14.6 [14.6,16.7]	4/0/0	5/0	7	10.3 [10.3,10.3]

A.2 TODO: LARGER-SCALE TRAINING RESULTS

TODO: Insert larger-scale training details here.

A.3 TODO: COMPONENT ABLATIONS

TODO: Insert component ablation details here.

B RATIONAL LATENT BASIS CALCULUS

The RLB summaries and the reciprocal matrix-pair coordinate follow from the scalar response and the RMS-normalized group map. We suppress layer and group indices throughout this section.

B.1 POLE-FREE RATIONAL RESPONSE

Let $\phi(u)$ collect the nonconstant denominator features. The evaluated P5/Q4 response is

$$\begin{aligned} P(u) &:= \sum_{i=0}^5 a_i u^i, & \phi(u) &:= (|u|, u^2, |u|^3, u^4), \\ Q(u) &:= 1 + \sum_{j=1}^4 |b_j| \phi_j(u), & R(u) &:= \frac{P(u)}{Q(u)}. \end{aligned} \quad (26)$$

Because each denominator feature is nonnegative, $Q(u) \geq 1$ for every $u \in \mathbb{R}$; hence the response is pole-free on $\mathbb{R}$. Away from the kink at $u = 0$,

$$\begin{aligned} R'(u) &= \frac{P'(u)Q(u) - P(u)Q'(u)}{Q(u)^2}, \\ P'(u) &= \sum_{i=1}^5 i a_i u^{i-1}, & Q'(u) &= |b_1| \text{sign}(u) + 2|b_2|u + 3|b_3|u|u| + 4|b_4|u^3. \end{aligned} \quad (27)$$

The selected coefficient derivatives are

$$\frac{\partial R(u)}{\partial a_i} = \frac{u^i}{Q(u)}, \quad \frac{\partial R(u)}{\partial b_j} = -\frac{P(u)}{Q(u)^2} \text{sign}(b_j) \phi_j(u). \quad (28)$$

The implementation selects $\text{sign}(0) = 0$ for both $u = 0$ and $b_j = 0$. At these kinks, the formulas specify the backward derivative chosen by autodiff. For vector inputs, R and R' act coordinatewise. The response, derivative, and coefficient-activity summaries therefore originate from $R(u)$, $R'(u)$, and the features in equation 28, respectively. For the evaluated response, $\Theta_{l,g}^R = \{a_{l,g,\cdot}, b_{l,g,\cdot}\}$ in equation 12.

B.2 RMS-NORMALIZED GROUP DIFFERENTIAL

Let m be the group width and let $\epsilon_{\text{rms}} \geq 0$. For $\epsilon_{\text{rms}} > 0$ and $z \in \mathbb{R}^m$, or for $\epsilon_{\text{rms}} = 0$ and $z \neq 0$, define

$$r(z) = \sqrt{m^{-1} \|z\|_2^2 + \epsilon_{\text{rms}}}, \quad u(z) = \frac{z}{r(z)}, \quad h(z) = r(z)R(u(z)). \quad (29)$$

On the domain $r(z) > 0$, direct differentiation gives

$$\text{d}r = \frac{z^\top \text{d}z}{mr} = \frac{u^\top \text{d}z}{m}, \quad \text{d}u = \frac{1}{r} \left(I_m - \frac{uu^\top}{m} \right) \text{d}z. \quad (30)$$

The factor $I_m - uu^\top/m$ is the exact normalized-coordinate differential. It is the tangent projector to the RMS sphere when $\epsilon_{\text{rms}} = 0$. With a positive floor, its radial eigenvalue is $\epsilon_{\text{rms}}/r^2$. Away from scalar-response kinks, the group Jacobian is

$$J_h(z) := \frac{\partial h}{\partial z} = R(u) \frac{u^\top}{m} + \text{diag}(R'(u)) \left(I_m - \frac{uu^\top}{m} \right). \quad (31)$$

This decomposition identifies the scalar factors observed by the live summaries. Response RMS summarizes the radial response factor, and derivative RMS summarizes $\text{diag}(R'(u))$. The complete local Jacobian additionally depends on the normalized-coordinate factor and radial term; the matrix-role gradient also depends on the output matrix and upstream adjoint. The two RMS summaries therefore serve as role-specific scalar proxies.

B.3 POSITIVE PAIR COORDINATE AND FLOOR PERTURBATION

Set $\epsilon_{\text{rms}} = 0$ and write r_0, u_0, h_0 for the unfloored map. For $z \neq 0$ and $\lambda > 0$,

$$r_0(\lambda z) = \lambda r_0(z), \quad u_0(\lambda z) = u_0(z), \quad h_0(\lambda z) = \lambda h_0(z). \quad (32)$$

For a group contribution $y_{l,g} = B_{l,g} h(A_{l,g} x_l)$, with R fixed, $\epsilon_{\text{rms}} = 0$, and $A_{l,g} x_l \neq 0$, the transformation $(A_{l,g}, B_{l,g}) \mapsto (\lambda A_{l,g}, \lambda^{-1} B_{l,g})$ leaves $y_{l,g}$ unchanged. This is the positive pair orbit in equation 9.

The RMS floor perturbs the orbit through both the radius and normalized coordinate. Write

$$r_\epsilon(z) = \sqrt{m^{-1}\|z\|_2^2 + \epsilon_{\text{rms}}}, \quad u_\epsilon(z) = z/r_\epsilon(z), \quad h_\epsilon(z) = r_\epsilon(z)R(u_\epsilon(z)). \quad (33)$$

For $\rho^2 = m^{-1}\|z\|_2^2 > 0$ and $\delta = \epsilon_{\text{rms}}/\rho^2$, define

$$s_\epsilon(\lambda, z) := \frac{r_\epsilon(\lambda z)}{\lambda r_\epsilon(z)} = \sqrt{\frac{1 + \delta/\lambda^2}{1 + \delta}}, \quad \kappa(\lambda, z) := s_\epsilon(\lambda, z)^{-1}, \quad (34)$$

$$u_\epsilon(\lambda z) = \kappa(\lambda, z)u_\epsilon(z).$$

Fix $c > 0$, and let $\mathcal{I}_c$ contain every scalar coordinate on all segments joining $u_\epsilon(z)$ and $\kappa(\lambda, z)u_\epsilon(z)$ for $\lambda \in [e^{-c}, e^c]$ and all sufficiently small δ . Suppose that R is L_R -Lipschitz on $\mathcal{I}_c$. Reciprocal output compensation then gives, for every λ in this interval, the latent feature bound

$$\begin{aligned} \|\lambda^{-1}h_\epsilon(\lambda z) - h_\epsilon(z)\|_2 &\leq \left| \frac{r_\epsilon(\lambda z)}{\lambda} - r_\epsilon(z) \right| \|R(u_\epsilon(z))\|_2 \\ &\quad + \frac{r_\epsilon(\lambda z)}{\lambda} L_R |\kappa(\lambda, z) - 1| \|u_\epsilon(z)\|_2. \end{aligned} \quad (35)$$

After the pair map, the one-group output difference is $B_{l,g}[\lambda^{-1}h_\epsilon(\lambda z) - h_\epsilon(z)]$, where $B_{l,g}$ denotes the pre-rescale output block. Its norm is at most $\|B_{l,g}\|_{\text{op}}$ times the right-hand side of equation 35. For fixed c and fixed m , uniformly over $\lambda \in [e^{-c}, e^c]$ as $\delta \rightarrow 0$,

$$\begin{aligned} \sup_{\lambda \in [e^{-c}, e^c]} (|s_\epsilon(\lambda, z) - 1| + |\kappa(\lambda, z) - 1|) &= O_c(\delta), \\ \sup_{\lambda \in [e^{-c}, e^c]} \|\lambda^{-1}h_\epsilon(\lambda z) - h_\epsilon(z)\|_2 &= O_c \left(\sqrt{m} [\sup_{\mathcal{I}_c} |R| + L_R] \frac{\epsilon_{\text{rms}}}{\rho} \right). \end{aligned} \quad (36)$$

The first line follows directly from $|s_\epsilon - 1| \leq \frac{1}{2}\delta|\lambda^{-2} - 1|$ and $|\kappa - 1| = |s_\epsilon - 1|/s_\epsilon$. For radii separated from zero and bounded reciprocal moves, the error decays as $\epsilon_{\text{rms}}/\rho$; this rate motivates the clipped log-scale update in MatrixPolicy.

C OPTIMIZER DESIGN AND MATRIXPOLICY GEOMETRY

Let $\text{sg}(\cdot)$ denote stop-gradient. MatrixPolicy first forms detached role summaries, then rescales the current matrix gradients, applies the AdamW/Muon matrix branch, and finally moves the updated pair along the positive coordinate from Appendix B.3. Its effective order is

$$\begin{aligned} \nabla \mathcal{L} &\longrightarrow \text{sg}(q^{\text{in}}, q^{\text{out}}, q^R, \pi, \alpha, \hat{d}^{\text{live}}, \hat{o}^{\text{live}}) \longrightarrow (\tilde{\nabla} A, \tilde{\nabla} B) \\ &\longrightarrow (A^{\text{step}}, B^{\text{step}}) \longrightarrow \text{sg}(\hat{d}^{\text{probe}}, \hat{o}^{\text{probe}}, \gamma) \\ &\longrightarrow (e^\ell A^{\text{step}}, e^{-\ell} B^{\text{step}}). \end{aligned} \quad (37)$$

C.1 LOCAL MATRIX-ROLE GEOMETRY

For one token of one training example, let $\bar{y}_l = \nabla_{y_l} \mathcal{L}$ and $J_{l,g} = \partial h_{l,g} / \partial z_{l,g}$ from equation 31. Here x_l is the RLB input, $z_{l,g} = A_{l,g}x_l$, $h_{l,g} = h(z_{l,g})$, and $y_l = \sum_g B_{l,g}h_{l,g}$. The adjacent matrix gradients factor as

$$\nabla_{B_{l,g}} \mathcal{L} = \bar{y}_l h_{l,g}^\top, \quad \nabla_{A_{l,g}} \mathcal{L} = (J_{l,g}^\top B_{l,g}^\top \bar{y}_l) x_l^\top. \quad (38)$$

Batch and sequence gradients are sums or averages of these outer-product terms. The output-role gradient contains the realized feature $h_{l,g}$, whereas the input-role gradient contains the local Jacobian $J_{l,g}$. This factorization supports response RMS and derivative RMS as scalar role proxies. The full gradient retains the omitted radius, normalized-coordinate factor, matrix orientation, data distribution, and upstream adjoint.

The pressure summaries quantify relative gradient scale. For a pure gradient step $W^+ = W - \eta \mathcal{G}$, with $W \neq 0$ and $\eta \|\mathcal{G}\|_F \leq c \|W\|_F$ for fixed $c < 1$, Taylor expansion gives

$$\log \text{rms}(W^+) - \log \text{rms}(W) = -\eta \frac{\langle \mathcal{G}, W \rangle_F}{\|W\|_F^2} + O_c \left(\eta^2 \frac{\|\mathcal{G}\|_F^2}{\|W\|_F^2} \right), \quad (39)$$

while Cauchy–Schwarz yields

$$\left| \frac{\langle \mathcal{G}, W \rangle_F}{\|W\|_F^2} \right| \leq \frac{\|\mathcal{G}\|_F}{\|W\|_F} = \frac{\text{rms}(\mathcal{G})}{\text{rms}(W)}. \quad (40)$$

Thus $\text{rms}(\mathcal{G})/\text{rms}(W)$ bounds the magnitude of the first-order log-RMS change for a pure gradient step. The implemented p^{in} and p^{out} are floored versions of this nonnegative bound, computed from post-global-clipping gradients before the group policy. They summarize unsigned pre-policy gradient scale; AdamW and Muon subsequently construct signed, state-dependent updates. Their EMA contrast $\pi_{l,g}$ identifies the role with larger recent relative gradient scale. The activity contrast $\alpha_{l,g}$ compares raw rational-coefficient gradient energy with the geometric mean of the matrix-pressure memories and supplies a calibrated policy signal for relative coefficient activity.

In the unfloored fixed-response map, the positive pair orbit also satisfies

$$\left. \frac{d}{ds} \mathcal{L}(e^s A_{l,g}, e^{-s} B_{l,g}) \right|_{s=0} = \langle \nabla A_{l,g}, A_{l,g} \rangle_F - \langle \nabla B_{l,g}, B_{l,g} \rangle_F = 0. \quad (41)$$

The final control follows this zero-loss tangent in the idealized map to alter relative matrix pressure.

C.2 CENTERED GROUP CONTROL

The first control is most transparent in log space. Define the layerwise mean log gains

$$\bar{d}_l = \frac{1}{G} \sum_{j=1}^G \log \hat{d}_{l,j}^{\text{live}}, \quad \bar{o}_l = \frac{1}{G} \sum_{j=1}^G \log \hat{o}_{l,j}^{\text{live}}. \quad (42)$$

The gain factors in equation 18 obey

$$\log c_{l,g,\text{in}}^{\text{gain}} = -0.20\phi_t(\log \hat{d}_{l,g}^{\text{live}} - \bar{d}_l), \quad \log c_{l,g,\text{out}}^{\text{gain}} = -0.20\phi_t(\log \hat{o}_{l,g}^{\text{live}} - \bar{o}_l). \quad (43)$$

Above-geometric-mean derivative or response gain therefore reduces the corresponding role multiplier. The pressure factors act with opposite signs,

$$\log c_{l,g,\text{in}}^{\text{pressure}} = -0.10\phi_t \tilde{\pi}_{l,g}, \quad \log c_{l,g,\text{out}}^{\text{pressure}} = +0.10\phi_t \tilde{\pi}_{l,g}, \quad (44)$$

so positive $\pi_{l,g}$ reduces the input-role gradient and increases the output-role gradient before recentering. The shared activity factor is

$$\log c_{l,g}^{\text{activity}} = -0.20\phi_t \text{ReLU}\left(\frac{\alpha_{l,g} - 0.05}{0.45}\right). \quad (45)$$

To distinguish the stages of the implementation, let

$$c_{l,g,\sigma}^{\text{pre}} := c_{l,g,\sigma}^{\text{gain}} c_{l,g,\sigma}^{\text{pressure}} c_{l,g}^{\text{activity}}, \quad \hat{c}_{l,g,\sigma} := \frac{c_{l,g,\sigma}^{\text{pre}}}{\text{geomean}_{j=1}^G c_{l,j,\sigma}^{\text{pre}}}, \quad c_{l,g,\sigma} := \text{clip}(\hat{c}_{l,g,\sigma}, 0.75, 1.35), \quad (46)$$

$$\frac{1}{G} \sum_{g=1}^G \log \hat{c}_{l,g,\sigma} = 0.$$

The zero-mean identity holds exactly before the final clip; the clip may shift the layerwise mean log multiplier. The gate preserves each group-gradient direction and multiplies its magnitude by a positive scalar before AdamW and Muon construct their state-dependent update directions.

C.3 TRANSIENT SPECTRAL NORMALIZATION

The role/depth factors in equation 20 are bounded scalar preconditioners. They preserve each matrix direction while changing the step magnitude assigned to the input and output maps across depth. AdamW then maintains entrywise moments (Kingma & Ba, 2015; Loshchilov & Hutter, 2019),

$$M \leftarrow \beta_1 M + (1 - \beta_1) G, \quad V \leftarrow \beta_2 V + (1 - \beta_2)(G \odot G), \quad (47)$$

$$D_{ij}^{\text{Adam}} = \frac{M_{ij}}{\sqrt{V_{ij}} + \epsilon},$$

where the display suppresses bias correction and decoupled weight decay. This normalization acts coordinatewise; the subsequent Muon branch controls the singular spectrum of the matrix direction.

Matrix spectra matter for the RLB pair because equation 38 gives single-example adjacent gradients as outer products; after averaging over batch and sequence positions, the training gradients are sums of such terms. Singular modes of B_l couple output-residual directions to latent features, and singular modes of A_l couple latent coordinates to input-residual directions. During early basis formation, a highly anisotropic singular spectrum concentrates movement in a narrow set of these coupled directions.

Let $H_{l,\sigma,t} = U\Sigma V^\top$ be the compact SVD of the tensor passed to Newton–Schulz after group-gradient scaling and Muon’s momentum/Nesterov construction. Its ideal polar direction is

$$\mathcal{P}(H_{l,\sigma,t}) = UV^\top, H = \sum_i \sigma_i u_i v_i^\top \mapsto \sum_i u_i v_i^\top \quad \text{on the nonzero singular subspace.} \quad (48)$$

The polar map preserves the row and column singular vectors, replaces every nonzero singular value by one, and retains the nonzero singular subspace of $H_{l,\sigma,t}$. The polar factor solves

$$\mathcal{P}(H) \in \arg \max_{\|D\|_2 \leq 1} \langle D, H \rangle_F, \quad \langle \mathcal{P}(H), H \rangle_F = \|H\|_*, \quad (49)$$

$$\mathcal{P}(aH) = \mathcal{P}(H) \quad (a > 0, H \neq 0).$$

The implementation uses $D_{l,\sigma,t}^\mu = \text{NS}_5(H_{l,\sigma,t})$, a finite Newton–Schulz approximation to the polar direction, together with match-RMS learning-rate adjustment (Liu et al., 2025).

The branch equation in equation 21 is sequential because AdamW and Muon keep separate states. Let $\mathcal{D}_{l,\sigma,t}^{\text{Adam}}$ and $\mathcal{D}_{l,\sigma,t}^\mu$ denote their complete first-order directions, including bias correction, decoupled weight decay, and, for Muon, the finite Newton–Schulz and match-RMS adjustments. The sequential step has expansion

$$\Delta M_{l,\sigma} = -\eta_t s_{l,\sigma}^{\text{Adam}} (1 - \mu_{l,\sigma,t}) \mathcal{D}_{l,\sigma,t}^{\text{Adam}} - \eta_t s_\mu \mu_{l,\sigma,t} \mathcal{D}_{l,\sigma,t}^\mu + O(\eta_t^2), \quad (50)$$

after the group control has set the role-specific gradient scales. Flattening the nonzero singular values gives weaker represented modes more relative movement within the existing singular subspace. MatrixPolicy confines this intervention to the early window in equation 20; AdamW and reciprocal pair balancing remain after the Muon fraction reaches zero.

The Muon gate in equation 20 is monotone in the two layer summaries. Away from clipping boundaries and the ReLU kink, the derivatives of $\log \psi_{l,t}$ with respect to $\Psi_{l,t}$ and $\Omega_{l,t}$ are -0.30 and -0.65 inside the ψ clipping interval. Thus the Muon fraction is largest when role pressure and rational-coefficient activity are both small. Large $|\pi_{l,g}|$ marks unequal input/output pressure, and large $\alpha_{l,g}$ marks rational coefficients moving strongly relative to the matrix pair. These are precisely the regimes where the spectrum-normalized substep is attenuated.

C.4 RECIPROCAL PAIR BALANCING

After the descent branch, MatrixPolicy applies the reciprocal reparameterization

$$A_{l,g}^+ = e^{\ell_{l,g,t}} A_{l,g}, \quad B_{l,g}^+ = e^{-\ell_{l,g,t}} B_{l,g}, \quad (51)$$

every five optimizer steps. The construction follows the positive coordinate in Appendix B.3. With the rational response held fixed, the unfloored map satisfies $h_{l,g}(\lambda A_{l,g} x) = \lambda h_{l,g}(A_{l,g} x)$ for $\lambda > 0$, so replacing $(A_{l,g}, B_{l,g})$ by $(\lambda A_{l,g}, \lambda^{-1} B_{l,g})$ leaves $B_{l,g} h_{l,g}$ unchanged. The floor-error bound in equation 35 quantifies the departure from this identity.

Equivalent unfloored representatives have different relative gradient pressures. Let

$$p_{l,g}^A = \frac{\text{rms}(\nabla A_{l,g})}{\text{rms}(A_{l,g})}, \quad p_{l,g}^B = \frac{\text{rms}(\nabla B_{l,g})}{\text{rms}(B_{l,g})}. \quad (52)$$

In the fixed-response, unfloored map, $J_h(e^\ell z) = J_h(z)$ almost everywhere. Evaluating the instantaneous gradients at the transformed representative gives

$$\begin{aligned} \nabla_{A_{l,g}^+} \mathcal{L} &= e^{-\ell_{l,g,t}} \nabla_{A_{l,g}} \mathcal{L}, & \nabla_{B_{l,g}^+} \mathcal{L} &= e^{\ell_{l,g,t}} \nabla_{B_{l,g}} \mathcal{L}, \\ p_{l,g}^{A,+} &= e^{-2\ell_{l,g,t}} p_{l,g}^A, & p_{l,g}^{B,+} &= e^{2\ell_{l,g,t}} p_{l,g}^B. \end{aligned} \quad (53)$$

The transformation law concerns freshly evaluated pressures; the EMA memories retain their temporal role in policy control. Positive ℓ reduces input-side relative pressure and increases output-side relative pressure. The $+0.25\pi_{l,g}$ term in equation 22 therefore raises $\tau_{l,g}$ when the input role has larger recent pressure, selecting the corrective sign in equation 53.

The curve-shape target follows from a separable role-pressure surrogate. After holding data- and orientation-dependent factors fixed within one group, write

$$\tilde{p}_{l,g}^A(\gamma) = \hat{d}_{l,g}^{\text{probe}} e^{-\gamma}, \quad \tilde{p}_{l,g}^B(\gamma) = \hat{o}_{l,g}^{\text{probe}} e^{\gamma}. \quad (54)$$

Their unique equality point is

$$\gamma_{l,g}^* = \frac{1}{2} \left(\log \hat{d}_{l,g}^{\text{probe}} - \log \hat{o}_{l,g}^{\text{probe}} \right), \quad (55)$$

which supplies the first term of $\tau_{l,g}$. This equality is a balance coordinate for the scalar proxy. Full data-dependent matrix gradients retain the orientation and data factors held fixed by the surrogate.

For nonzero matrices and unfloored RMS values,

$$\begin{aligned} \gamma_{l,g} &= \log \text{rms}(A_{l,g}) - \log \text{rms}(B_{l,g}), \\ \gamma_{l,g}^+ &= \gamma_{l,g} + 2\ell_{l,g,t}. \end{aligned} \quad (56)$$

Substituting the unclipped log step from equation 23 gives

$$\begin{aligned} \zeta_{l,g,t} &= s_{l,t} \chi_{l,g}^{\text{rs}}, \\ \gamma_{l,g}^+ &= (1 - \zeta_{l,g,t}) \gamma_{l,g} + \zeta_{l,g,t} \tau_{l,g}. \end{aligned} \quad (57)$$

Since $\delta_l \in [0, 1]$, the evaluated depth factor lies in $[0.925, 1.075]$. Hence $0 \leq s_{l,t} \leq 0.5375$ and $\chi_{l,g}^{\text{rs}} \leq e^{0.125}$, so $0 \leq \zeta_{l,g,t} < 0.61$. The idealized unclipped coordinate is therefore a strict contraction toward $\tau_{l,g}$ whenever $\zeta_{l,g,t} > 0$. The 0.030 log-step clip bounds its magnitude and shortens moves that exceed the bound. With floored RMS values, $\gamma^+ = \gamma + 2\ell$ is approximate, and the implementation relies on the local bound in Appendix B.3.

The activity factor has the opposite sign in this speed control. Large $\alpha_{l,g}$ damps gradient-driven matrix motion in equation 18 and equation 20; here $\chi_{l,g}^{\text{rs}} = \exp(0.10\tilde{\alpha}_{l,g})$ controls the speed of the post-step pair move, while the group and Muon controls determine gradient-driven matrix motion.